

**AI-BASED MULTICRITERIA OPTIMIZATION**

CANADA 

Course Presentation

Lecture 1 – Tuesday May 15, 2018

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Outline

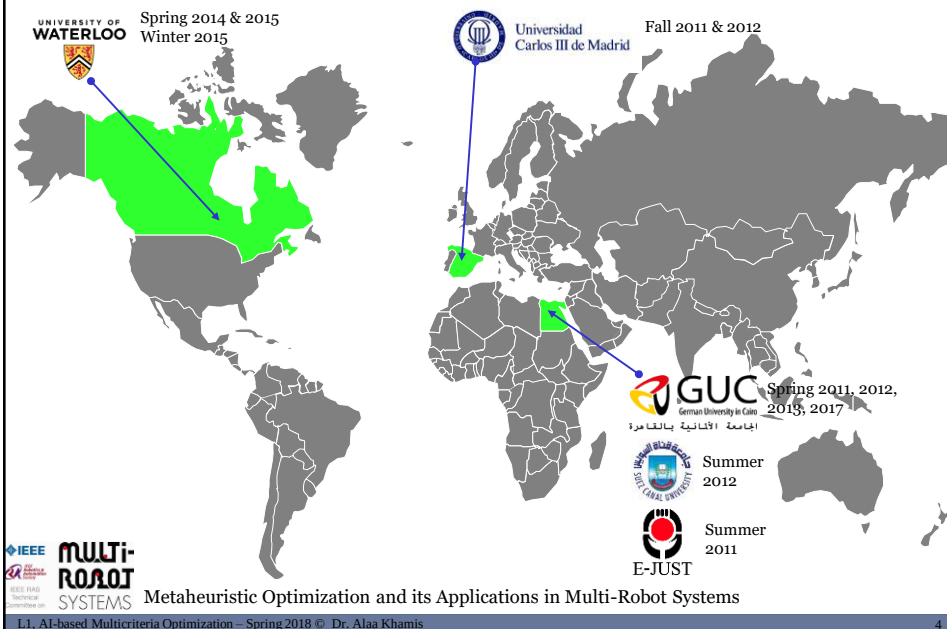
- Course Description
- Course Topics
- Course Outline
- Teaching Methodology
- Course Project
- Resources

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Outline

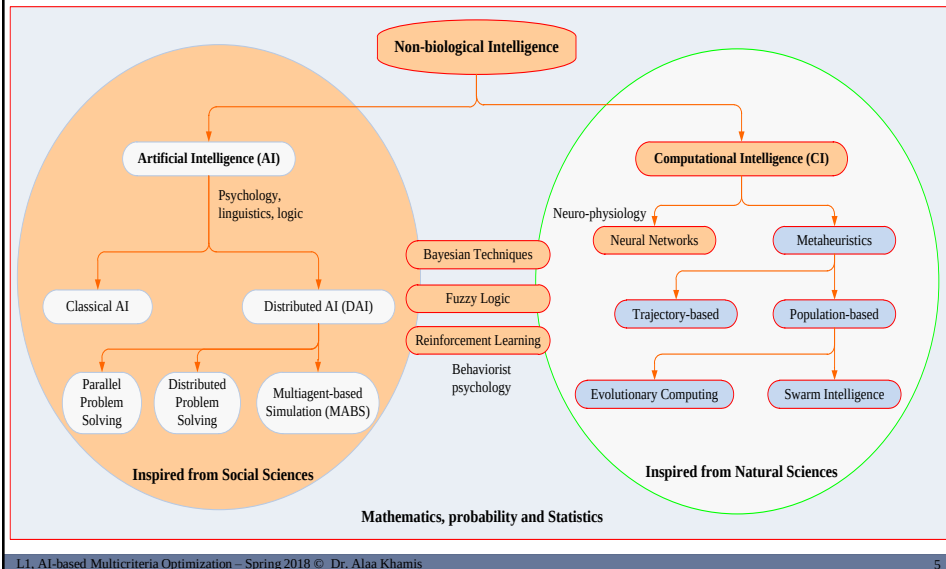
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Course Description



Course Description

Machine Intelligence = Non-biological Intelligence



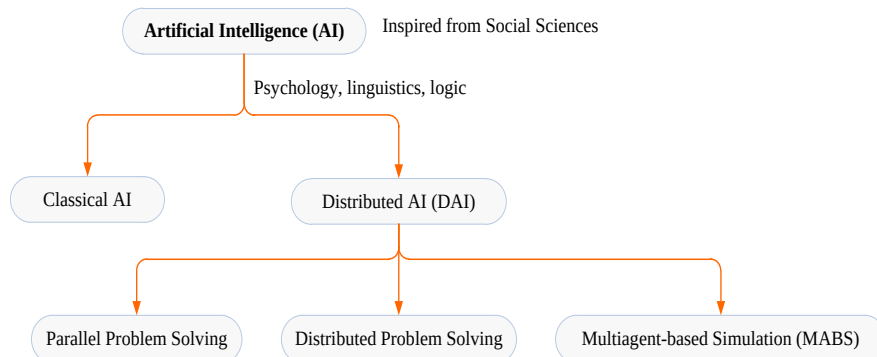
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Course Description

• Artificial Intelligence (AI):

AI deals with helping machines find solutions to **complex problems** in a more **human-like fashion**. This usually involves borrowing characteristics from human intelligence, and applying them as algorithms in a computer friendly way [3].



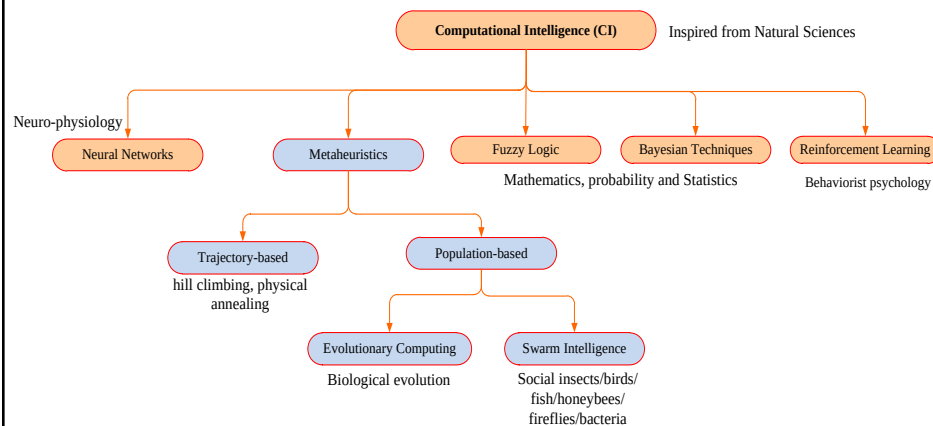
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Course Description

• Computational Intelligence (CI)

CI is defined as a part of computer science devoted to solution of non-algorithmizable problems.



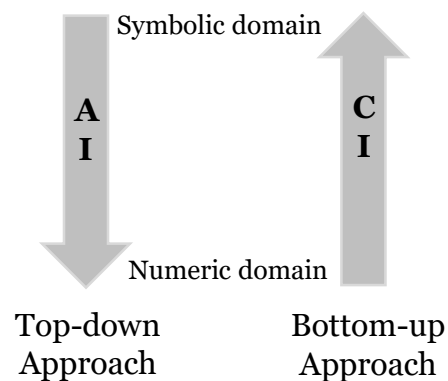
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Course Description

• AI vs. CI:

CI is characterized as bottom-up (working on numeric data to infer symbols), while AI used to work in the symbolic domain (top-down).



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Course Description

- **Heuristics:** 8-Puzzle/sliding-block puzzle/tile-puzzle Problem

1	4	2
3	7	5
6		8

Current State



	1	2
3	4	5
6	7	8

Goal State

- **States:** location of blank and location of the 8 tiles
- **Operator (successor):** blank moves left, right, up and down
- **Goal:** match the state given by the Goal state
- **Solution/Path:** sequence through state space;
- **Stopping Criteria:** ordered puzzle;
- **Evaluation Criteria:** number of steps or path cost (the length of path).

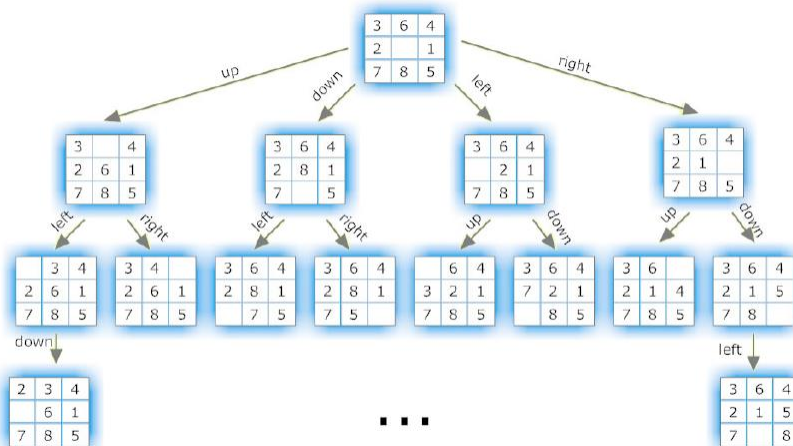
[Ref.: Crina Grosan and Ajith Abraham. *Intelligent Systems: A Modern Approach*. Springer, 2011]

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Course Description

- **WSP: 8-Puzzle Problem**



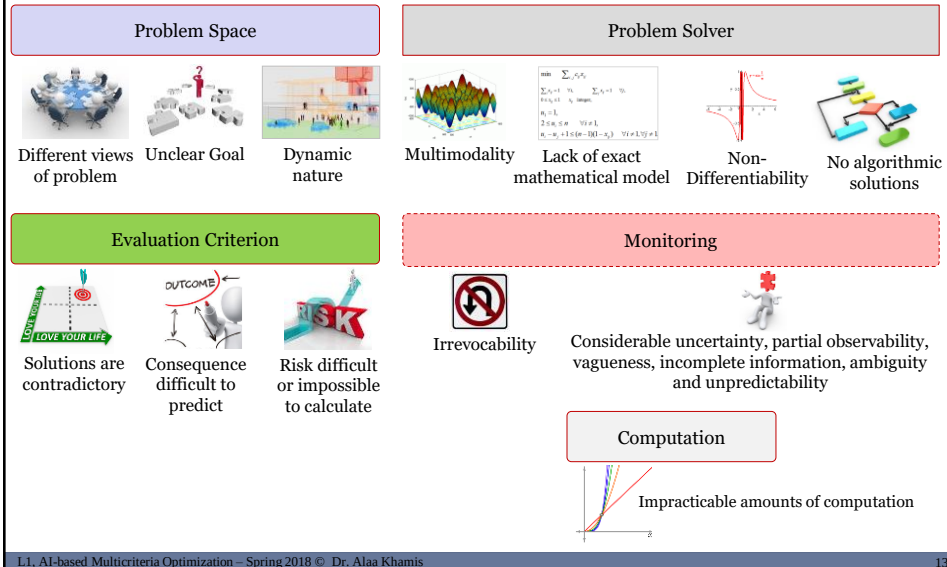
Using **combinatorial search**, the 8-puzzle can be solved in 31 moves or fewer.

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Course Description

• Ill-structured Problems (ISP)

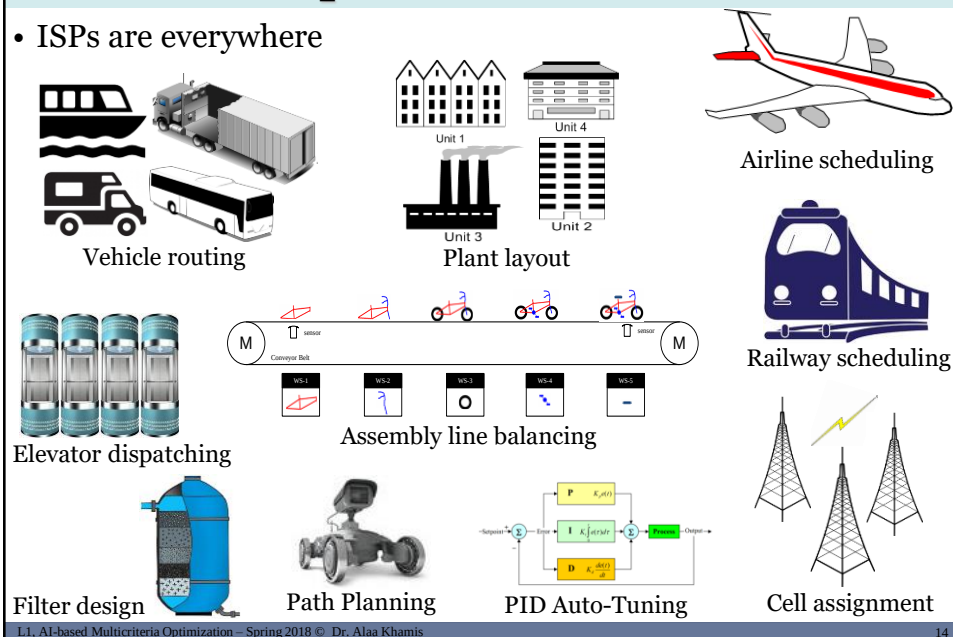


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Course Description

• ISPs are everywhere

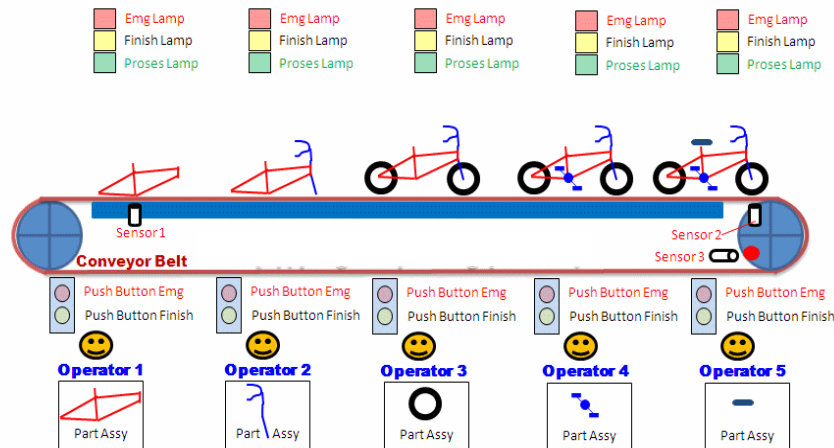


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Course Description

• ISP: Assembly Line Balancing



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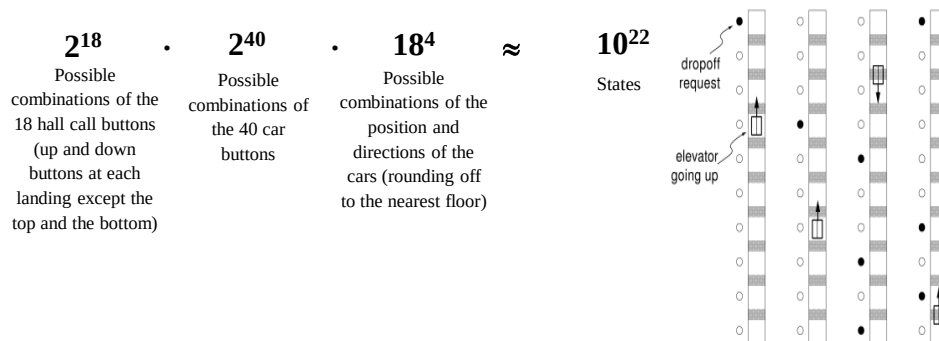
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Course Description

• ISP: Elevator Dispatching

Elevator Dispatching is a stochastic optimal control problem of economic importance that is too large to solve by classical techniques such as dynamic programming.

A four-elevator, ten-floor system has over 10^{22} states.



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Course Description

• WSP but practically ISP: TSP

- ◇ Given n cities,
- ◇ A travelling salesman must visit the n cities and return home, making a loop (roundtrip).
- ◇ He would like to travel in the most efficient way (cheapest way or shortest distance or some other criterion).



• Search Space

Search space is BIG:

for 21 cities there are $21! \approx 51,090,942,171,709,440,000$ **possible** tours. This is NP-Hard problem.

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Course Description

• WSP but practically ISP: TSP

TSP

Symmetric

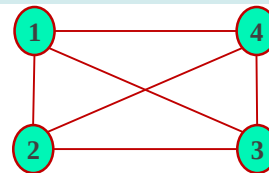
In the symmetric TSP, the **distance between two cities is the same in each opposite direction**, forming an **undirected graph**. This symmetry halves the number of possible solutions.

$$\# \text{routes} = 4! / 2 = 12$$

Asymmetric

In the asymmetric TSP, **paths may not exist in both directions** or the **distances might be different**, forming a **directed graph**. Traffic collisions, one-way streets, and airfares for cities with different departure and arrival fees are examples of how this symmetry could break down.

$$\# \text{routes} = 4! = 24$$



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Course Description

TSP Applications

TSP is used as a **platform** for the study of general methods that can be applied to a wide range of **discrete optimization** problems

Microchips manufacturing: manufacture of a circuit board, it is important to determine the best order in which a laser will drill thousands of holes.

Assignment of routes for planes of a specified fleet

Permutation flow shop scheduling problem (PFSP): consists of finding a sequence for the jobs that minimizes the makespan; that is, the overall time to complete the schedule.

Transportation and Logistics: problem of arranging school bus routes to pick up the children in a school district, transportation of farming equipment from one location to another to test soil or scheduling of service calls at cable firms, the delivery of meals to homebound persons, the scheduling of stacker cranes in warehouses, the routing of trucks for parcel post pickup, and a host of others.

Overhauling of gas turbine engines

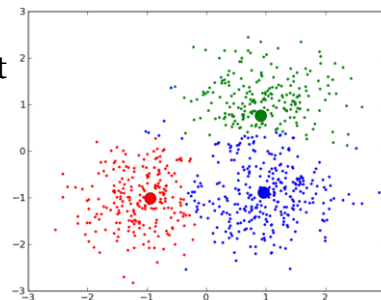
For more: http://iris.gmu.edu/~khoffman/papers/trav_salesman.html

Course Description

• WSP but practically ISP

The Clustering Problem

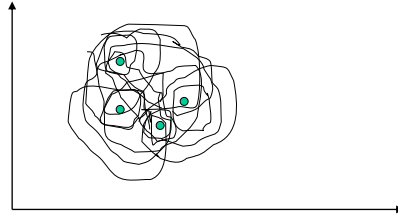
- ◇ Given n objects, group them in c groups (clusters) in such a way that all objects in a single group have a “natural” relation to one another, and objects not in the same group are somehow different.



Course Description

• WSP but practically ISP

The Clustering Problem



Course Description

• WSP but practically ISP

The Clustering Problem: Exponential Growth

$$N(n, c) = \left(\frac{1}{c!} \right) \sum_{m=1}^c (-1)^{c-m} \binom{c}{m} m^n$$

[See [partition theory](#)]

- ◇ Example: If ***n***=50 and ***c***=4, the number is **5.3x10²⁸**
- ◇ If ***n*** is increase to **100**, the number becomes **6.7x10⁵⁸**.
- ◇ So enumerating all possible partitions for large problems is practically not feasible

Course Description

• Heuristics

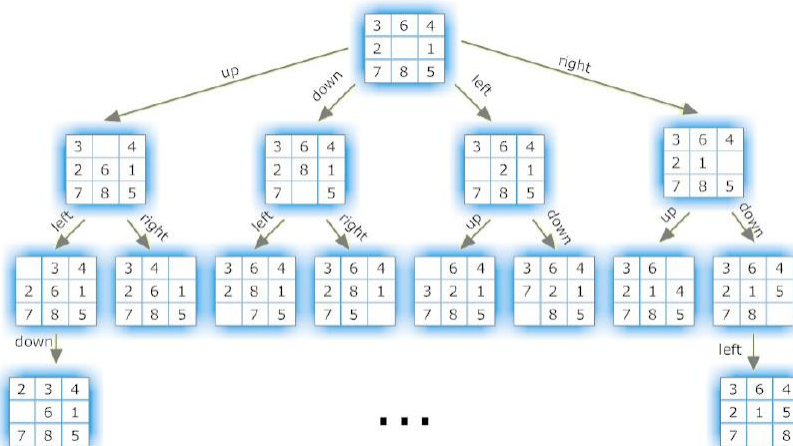
Heuristics is a solution strategy or rules by **trial-and-error** to produce acceptable (**optimal or near-optimal**) solutions to a complex problem in a reasonably practical time.

We use heuristics to **avoid or moderate** the **combinatorial explosion** of the classical algorithms.

Examples: FIFO, LIFO, earliest due date first, largest processing time first, shortest distance first, etc.

Course Description

• Heuristics: 8-Puzzle/sliding-block puzzle/tile-puzzle Problem



The 8-puzzle can be solved in 31 moves or fewer.

Course Description

- **Heuristics: 8-Puzzle/sliding-block puzzle/tile-puzzle Problem**

1	4	2
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Current State

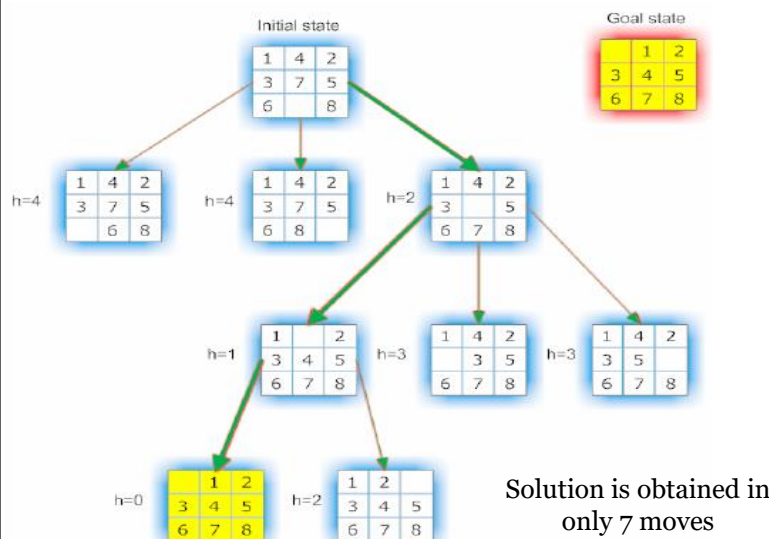
	1	2
3	4	5
6	7	8

Goal State

- ◇ We can define the following heuristics :
- ◇ $h_1(n)$ = the number of **misplaced tiles** in the current state relative to the goal state.
- ◇ In this example, $h_1(n) = 4$ because tiles 1, 4, 7 and blank are out of place. Obviously, lower values for h_1 are preferred.

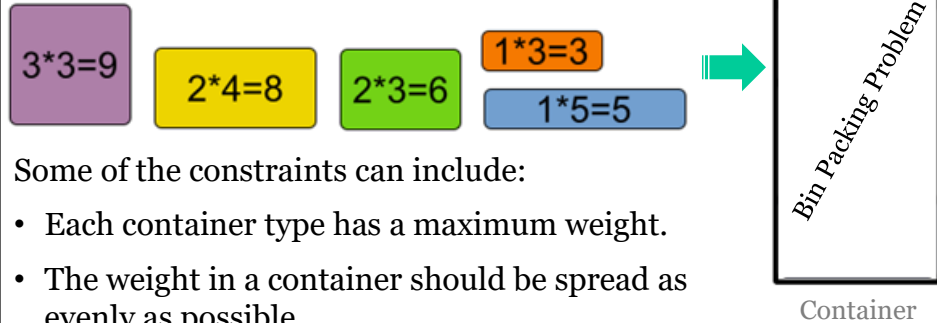
Course Description

- **Heuristics: 8-Puzzle Problem**



Course Description

• Heuristics: Bin Packing Problem



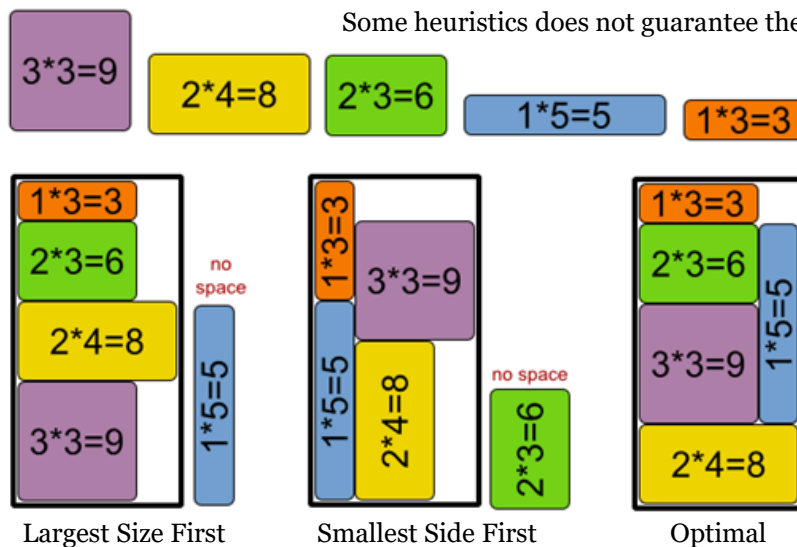
Some of the constraints can include:

- Each container type has a maximum weight.
- The weight in a container should be spread as evenly as possible.
- Different container types have different prices: minimize the overall cost.
- Not all items are added or removed from the container at the same time: minimize loading and unloading times.

Course Description

• Heuristics: Bin Packing Problem

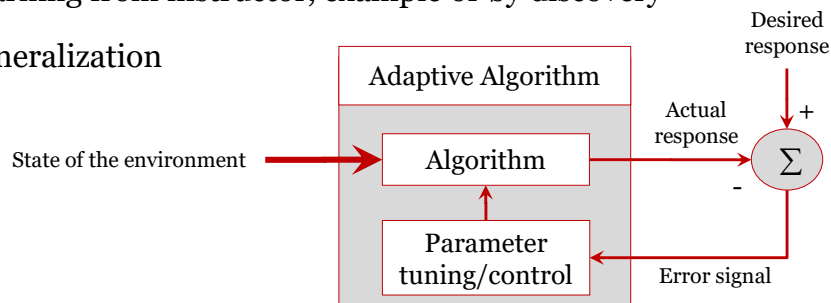
Some heuristics does not guarantee the optimality



Course Description

• Adaptive Algorithms

- ◇ To adjust to new or different situations
- ◇ Ability to improve behaviour
- ◇ Evolution
- ◇ Learning from instructor, example or by discovery
- ◇ Generalization



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Course Description

• Adaptive Algorithms: Types of adaptation

- ◇ **Deterministic:** when the control parameter is changed according to some deterministic update rule without taking into account any information from the search algorithm,
- ◇ **Adaptive:** when the update rule takes information from the search algorithm and changes the control parameter accordingly,
- ◇ **Self-adaptive:** when the update rule itself is being adapted.

[Ref.: A. E. Eiben, R. Hinterding and Z. Michalewicz. "Parameter Control in Evolutionary Algorithms". IEEE Transactions on Evolutionary Computing, vol. 3, issue 2, pp. 124-141, 1999.]

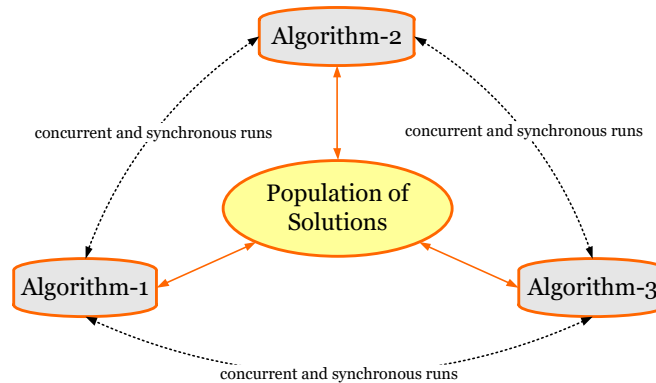
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Course Description

• Cooperative Algorithms

- ◇ A group to **solve a joint problem** or perform a common task(s) based on sharing the responsibility for **reaching the goal**.



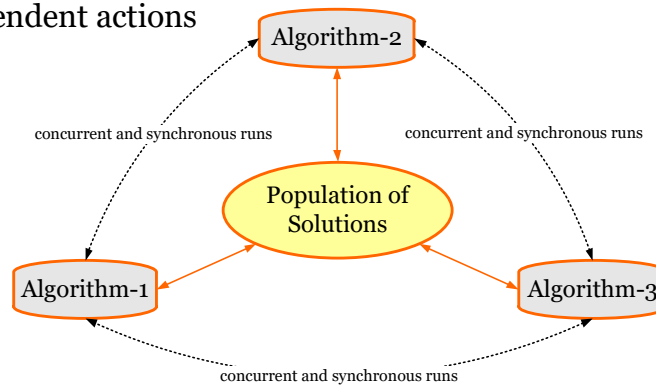
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Course Description

• Cooperative Algorithms

- ◇ Direct cooperation
- ◇ Indirect/Stigmergic cooperation
- ◇ Independent actions



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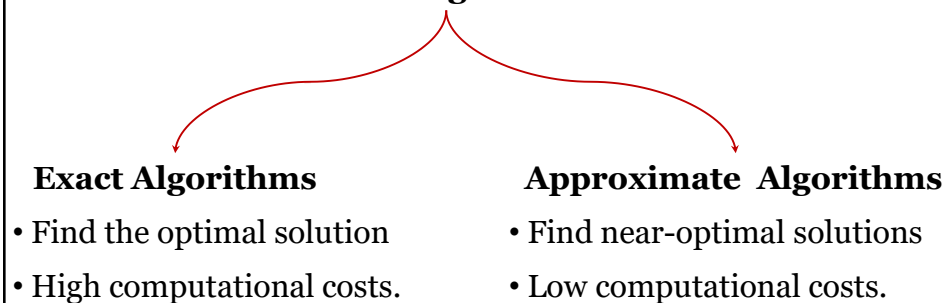
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Outline

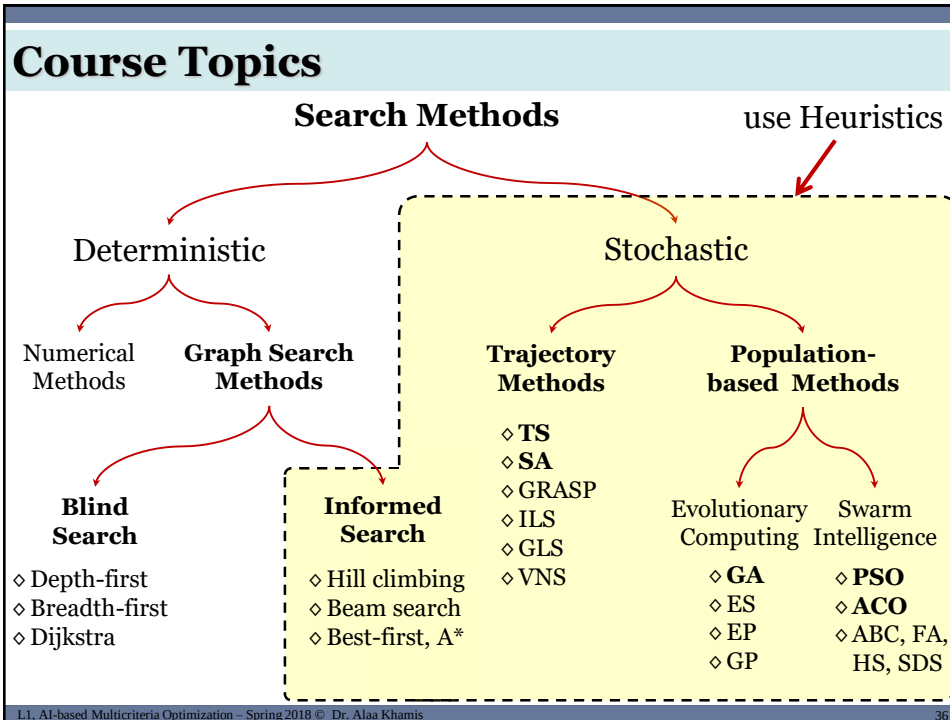
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Course Topics

Search Algorithms

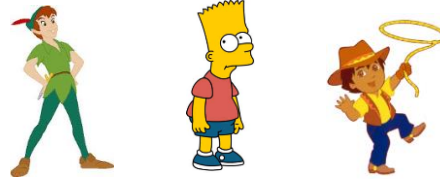


In most of the real applications, **quickly finding near-optimal** solution is better than **spending large amount of time** in search for optimal solution.



Course Topics

- ◇ Among your friends you have agreed on **some sharing mechanism** so that all who have taken part in the search will be rewarded, but with the person who found the treasure getting a **higher reward** than all others, and the rest being rewarded based on **distance from the treasure** at the time when the first one finds the treasure.

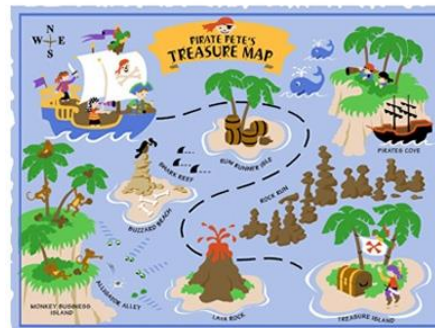
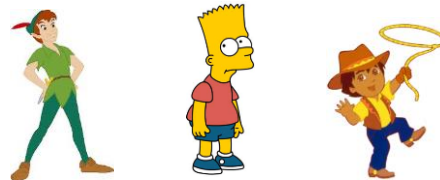


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Course Topics

- ◇ Each one in the group has a **metal detector** and can communicate the strength of the signal and his **current location** to the nearest neighbors.
- ◇ Each person therefore knows whether one of his neighbors is nearer to the treasure than he is.



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Course Topics

What actions will you take?

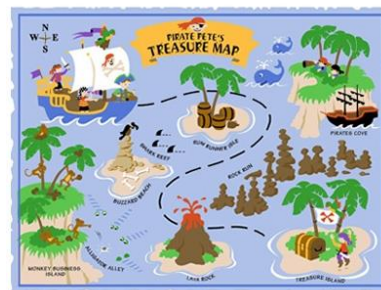


1. Ignore your friends, and search for the treasure **without any information** that your friends may provide. In this case, if you find the treasure, it is all yours. However, if you do not find it first, you get nothing.

2. Make use of the information that you perceive from your neighboring friends, and move in the direction of your closest friend with the strongest signal. By making use of local information, and acting upon it, you **increase your chances of finding the treasure**, or at least maximizing your reward.

Course Topics

◇ In one extreme, suppose we are blind-folded without any guidance, so the search process is essentially a pure random search (**blind search**), which is usually not efficient as we can expect.



Course Topics

- ◇ In another extreme, if we are told the treasure is placed at the **highest peak of a known region**, we will then directly climb up to the **steepest cliff** and try to reach to the highest peak, and this scenario corresponds to the classical **hill-climbing** techniques (**informed search**).



Course Topics

- ◇ If you are **not blind-folded**, and you **do not know where to look**.
- ◇ It is **a silly idea to search every single square inch** of an extremely large hilly region so as to find the treasure.
- ◇ In most cases, you will do **some random walking while looking for some hints**; we look at some places almost randomly, then move to another plausible place, then another and so on.
- ◇ Such **random walking** is a main characteristic of **modern meta-heuristic search algorithms**.

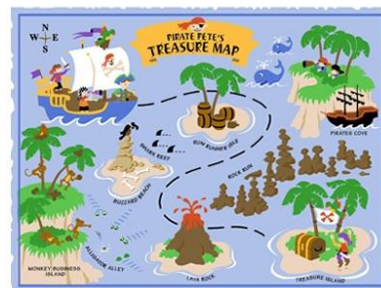
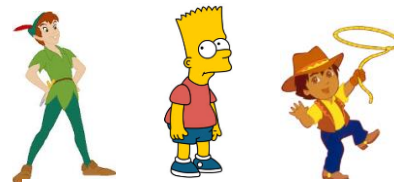
Course Topics

- ◇ You can do the treasure-hunting **alone**, so the whole path is a **trajectory-based search**
- ◇ Tabu search and simulated annealing are trajectory-based search algorithms.



Course Topics

- ◇ Alternatively, we can ask a **group of people** to do the hunting and share the information (and any treasure found), and
- ◇ This scenario uses the so-called **swarm intelligence** and corresponds to the ant colony optimization and particle swarm optimization.



Course Topics

- ◇ Some hunters are better than others. You can only keep the better hunters and recruit new ones;
- ◇ This is similar to the **genetic algorithms or evolutionary algorithms** where the search agents are improving.
- ◇ In fact, almost all modern metaheuristic algorithms, we try to use the best solutions or agents, and randomize (or replace) the not-so-good ones, while evaluating each individual's competence (fitness) in combination with the system history (use of memory). In such a balanced way, we intend to design better and more efficient optimization algorithms.



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Course Topics

Introductory Concepts

- Non-biological Intelligence and Computational Problems
- Optimization Theory
- Graph-search Algorithms
- Metaheuristics

Trajectory-based Metaheuristic Algorithms

- Tabu Search
- Simulated Annealing

Population-based Metaheuristic Algorithms

- P-Metaheuristics
- Evolutionary Computation
- Genetic Algorithms
- Swarm Intelligence
- Particle Swarm Optimization
- Ant Colony Optimization

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Course Topics

Self-study Algorithms

- Bat algorithm
- Firefly algorithm
- Artificial bee colony (ABC) algorithm
- Grey wolf optimizer
- Ant lion optimizer
- Intelligent Water Drops (IWD)
- River formation dynamics (RFD)
- Stochastic Diffusion Search (SDS)
- Harmony Search (HS) algorithm

Outline

- Course Description
- Course Topics
- **Course Outline**
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- Course Project
- Resources

Course Outline			
ID	Date	Lectures	Assignments
1	May 15	Course Presentation and Introduction	
2	May 22	Graph Search Algorithms	
3	May 29	Tabu Search	
4	June 5	Simulated Annealing	
5	June 12	Genetic Algorithms	Assignment-1 due date
6	June 19	Particle Swarm Optimization (PSO)	
7	June 26	Ant Colony Optimization (ACO)	Assignment-2 due date
8	July 17	Project Presentation	

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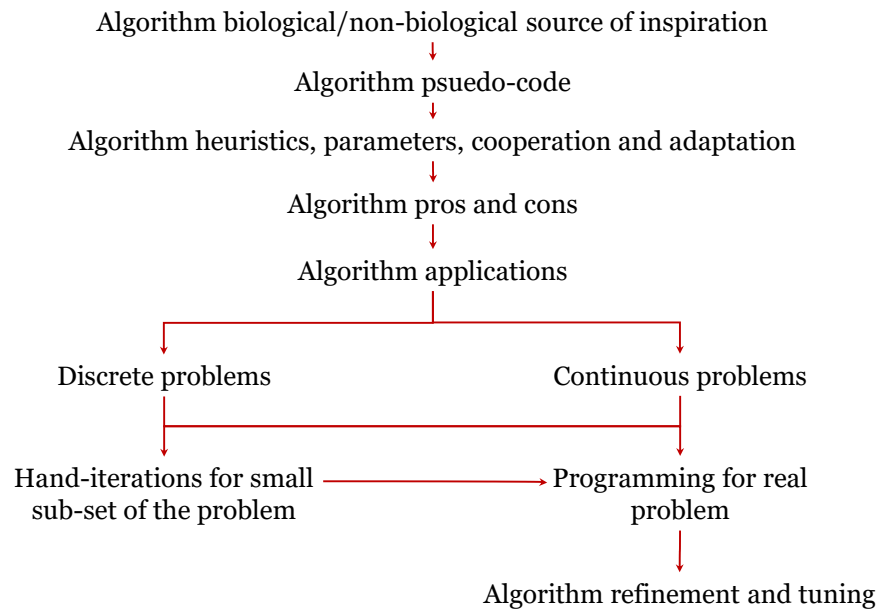
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Outline
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Teaching Methodology



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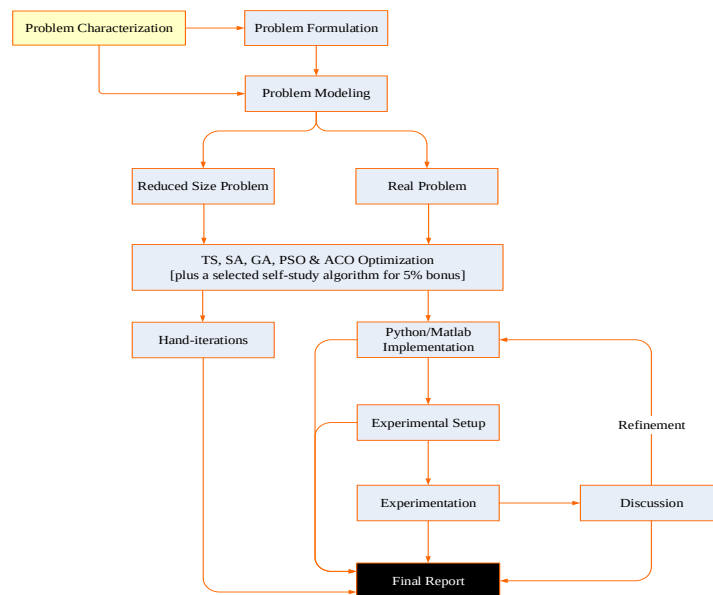
Course Project

- **ATW Projects:** Optimization is everywhere and metaheuristics can be applied in any ATW project.
- **Vehicle Projects:** Vehicle Suspension Design, Anti-lock Braking System, Auto-tuning of a PID (Proportional-Integral-Derivative) Controller, Autonomous Vehicle Path Planning Problem, Dispatching Problem, Test Automation, The Capacitated Vehicle Routing Problem, Autonomous Bus Routing Problem, Dynamic Location Problem, Combinatorial Optimization in Chip Design, Assembly Line Balancing Problem, etc.
- **Autonomous Vehicle Fleet Projects:** Autonomous Vehicle Fleet Routing Problem, Ride Sharing Problem, Timber Transport Vehicle Routing Problem (TTVRP), Driverless Shuttle Scheduling Problem, Autonomous Vehicle Control, Multiple Targets Clustering Problem, Optimal Sensor Placement, Static Waypoint Ordering Problem of Autonomous Vehicles, Automated People Mover (APM) Timetable Problem, Crew Scheduling, Complex Task Allocation Problem, Communication Relay for Autonomous Vehicles, Trajectory Planning and Assignment in Autonomous Vehicles, Adaptive Autonomous Vehicle Coverage Control, etc.

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Course Project



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Course Resources

- **Course Website**

Lecture slides, reading material, assignments, sample code will be posted on course website:

<https://gmweb.gm.com/sites/optimization>

Course Resources

• Textbook (Optional)

- ◇ A. Engelbrecht. *Fundamentals of Computational Swarm Intelligence*. Wiley, 2006.
- ◇ Xin-SheYang. *Engineering Optimization: An Introduction with Metaheuristic Applications*. A JOHN WILEY & SONS, INC., 2010.
- ◇ Singiresu S. Rao. *Engineering Optimization: Theory and Practice*. John Wiley & Sons, INC., 2009.
- ◇ C. Revees. *Modern heuristic techniques for combinatorial problems*. Halsted Press, New York, 1993.

Questions?